



UNIVERSITY OF MANITOBA  
Department of Mathematics

COURSE: *Math 1210*  
DATE: *May 26, 2023*  
EXAM: *Term test 2*

EXAMINER: *Romain Branchereau*  
TIME: *50 Minutes*

First name (please write as legibly as possible within the boxes)

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**Grading table**

|           |   |   |    |   |   |       |
|-----------|---|---|----|---|---|-------|
| Question: | 1 | 2 | 3  | 4 | 5 | Total |
| Points:   | 7 | 5 | 12 | 5 | 6 | 35    |
| Score:    |   |   |    |   |   |       |

**Instructions**

- Answer all questions on the exam paper in the space provided beneath the question. Unless otherwise specified, your answer must be justified. Unjustified answers will receive little or no credit.
- If you need more space, continue on the back of the same page, clearly indicating that your work is to be continued.
- The value of each question is indicated beside the statement of the question. The total value of all questions is 35 points.
- No texts, notes, or other aids are permitted. There are no calculators, cellphones, or any other electronic device permitted.
- Do not remove staples or any pages from the exam except the last blank page.
- If you have any questions, feel free to ask me.
- Good Luck!



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#1 Page 2 of 14



1. Fill in the blank spaces. Justifications are not required for this question.

(a) (1 point) The exponential form of  $1 - i$  is \_\_\_\_\_.

(b) (1 point) The cartesian form of  $8e^{7i\pi/6}$  is \_\_\_\_\_.

(c) (1 point) The degree of the polynomial  $x^{11} + x^2 - x$  is \_\_\_\_\_.

(d) (1 point) Consider the polynomial  $P(x) = x^4(x + 3)^5(x - 3)^6$ .

i. (1 point) The degree of  $P(x)$  is \_\_\_\_\_.

ii. (1 point) The root  $-3$  is of multiplicity \_\_\_\_\_.

iii. (1 point) \_\_\_\_\_ is a root of multiplicity 4.



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#1 Page 4 of 14

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2. (5 points) Find all solutions to the equation

$$z^3 = 8e^{i\pi/6}.$$

Write the solutions in exponential form using the principal value of the argument.



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#1 Page 6 of 14

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3. Consider the polynomial  $P(x) = 3x^5 - 2x^4 + 3x^3 - 5x^2 + 5x - 7$ .
- (a) (2 points) Use Bounds theorem to find bounds on the roots.
- (b) (3 points) Use Descartes rule of signs to determine the number of possible positive and negative real roots.
- (c) (3 points) Use parts (a) and (b) to deduce that all the roots of the polynomial are contained in the interval  $(0, 7)$ .
- (d) (4 points) Use the Rational root theorem to give a list of possible rational roots. Reduce the list as much as possible by using the previous parts.



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test-2-ab1c2

#1 Page 8 of 14

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4. (5 points) Consider the polynomial  $P(x) = 2x^{77} - m + 1$  where  $m$  is a real number. Find all values of  $m$  such that  $(x + 1)$  divides  $P(x)$ . Justify your answer.



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test-2-ab1c2

#1 Page 10 of 14

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5. (6 points) Consider the polynomial  $P(x) = x^3 + x^2 + 4x + 30$ . It is given to you that  $1 - 3i$  is a root of  $P(x)$ . List all the roots of  $P(x)$ .



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#1 Page 12 of 14

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#1 Page 13 of 14



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#1 Page 14 of 14

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